

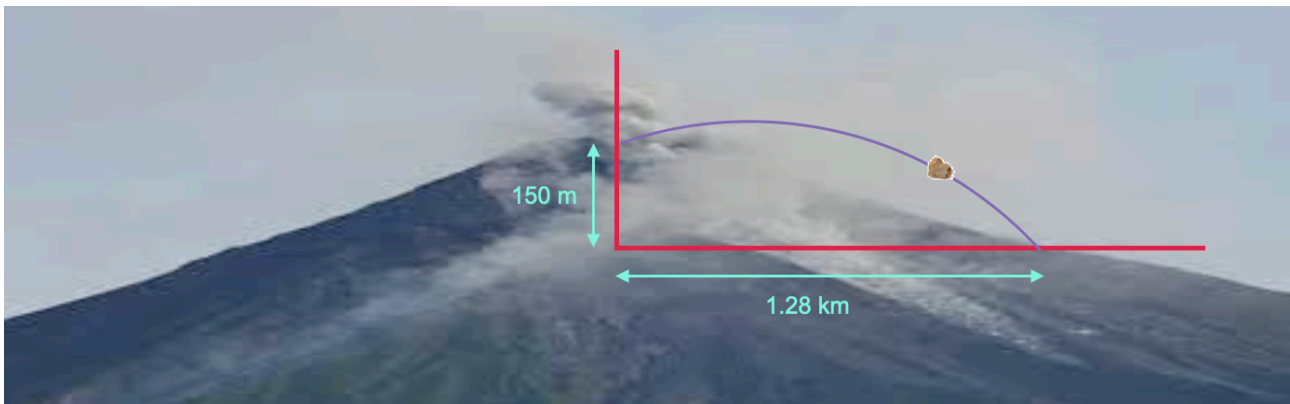
PHY 4830 Computational Astrophysics and Geophysics - Monsoon 2025

Assignment 2.

Issue: 16 Sep 2025

Due: 25 Sep 2025 11:59 pm

10 points



A volcano ejects a boulder of mass 85 kg from its summit, which after 10 seconds impacts the side of the mound at a location that is 150 m below the summit and at a horizontal distance of 1.28 km from the ejection point (see figure above). Set up a differential equation to describe the trajectory of the boulder and solve it numerically by

- (a) A shooting method, and
- (b) A relaxation method

And compare their accuracy and efficiency.

From the numerical solutions, record the velocity of the boulder at the start and the end of the trajectory. Compare these values with those derived analytically.

Assume that the acceleration due to gravity is constant and of value 9.8 m/s^2 .